100. What is $8\int_1^2 f(x)dx$ equal to?

(a) $\ln(8\sqrt{e})$ **(b)** $\ln(4\sqrt{e})$
(c) $\ln 2$ **(d)** $\ln 2 - 1$

101. A bag contains 5 black and 4 white balls. A man selects two balls at random. What is the probability that both of these are of the same colour?

(a) $\frac{1}{6}$ **(b)** $\frac{5}{108}$ **(c)** $\frac{4}{9}$ **(d)** $\frac{5}{18}$

102. If a random variable (x) follows binomial distribution with mean 5 and variance 4 and $5^{23}P(X = 3) = \lambda 4^{\lambda}$, then what is the value of λ?
(a) 3 **(b)** 5 **(c)** 23 **(d)** 25

103. From data (–4, 1), (–1, 2), (2, 7) and (3, 1), the regression line of y on x is obtained as $y = a + bx$, then what is the value of $2a + 15b$?
(a) 6 **(b)** 11 **(c)** 17 **(d)** 21

104. Let $x + 2y + 1 = 0$ and $2x + 3y + 4 = 0$ are two lines of regression computed from some bivariate data. If θ is the acute angle between them, then what is the value of $488 \tan 3\theta$?
(a) 191 **(b)** 161 **(c)** 131 **(d)** 121

105. If two random variables X and Y are connected by relation $\frac{2X-3Y}{5X+4Y} = 4$ and X follows Binomial distribution with parameters $n = 10$ and $p = \frac{1}{2}$, then what is the variance of Y?

a) $\frac{810}{361}$ **(b)** $\frac{9}{19}$ **(c)** $\frac{21}{361}$ **(d)** $\frac{121}{361}$

106. If a, b, c are in HP, then what is $\frac{1}{b-a}+\frac{1}{b-c}$ equal to?

1. $\frac{2}{b}$ **2.** $\frac{1}{a}+\frac{1}{c}$

3. $\frac{1}{2}\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$

Select the correct answer using the code given below:
(a) 1 only **(b)** 1 and 2 only
(c) 3 only **(d)** 1, 2 and 3

107. An edible oil is sold at the rates 150, 200, 250, 300 rupees per litre in four consecutive years. Assuming that an equal amount of money is spent on oil by a family in every year during these years, what is the average price of oil in rupees (approximately) per litre?
(a) 210 **(b)** 220 **(c)** 225 **(d)** 240

108. If the letters of the word "TIRUPATI" are written down at random, then what is the probability that both Ts are always consecutive?

(a) $\frac{1}{2}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{1}{7}$ **(d)** $\frac{1}{14}$

109. Let $m = 77^n$. The index n is given a positive integral value at random. What is the probability that the value of m will have 1 in the units place?

(a) $\frac{1}{2}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{n}$

110. Three different numbers are selected at random from the first 15 natural numbers. What is the probability that the product of two of the numbers is equal to third number?

(a) $\frac{1}{91}$ **(b)** $\frac{2}{455}$ **(c)** $\frac{1}{65}$ **(d)** $\frac{6}{455}$

*Consider the following for the next **two (02)** items that follow:*

Let A and B be two events such that $P(A \cup B) \geq 0.75$ and $0.125 \leq P(A \cap B) \leq 0.375$.

111. What is the minimum value of $P(A) + P(B)$?
(a) 0.625 **(b)** 0.750 **(c)** 0.825 **(d)** 0.875

112. What is the maximum value of $P(A) + P(B)$?
(a) 0.75 **(b)** 1.125 **(c)** 1.375 **(d)** 1.625

*Consider the following for the next **two (02)** items that follow:*

A, B and C are three events such that $P(A) = 0.6, P(B) = 0.4, P(C) = 0.5, P(A \cup B) = 0.8$, $P(A \cap C) = 0.3$ and $P(A \cap B \cap C) = 0.2$ and $P(A \cup B \cup C) \geq 0.85$.

113. What is the minimum value of $P(B \cap C)$?
(a) 0.1 **(b)** 0.2 **(c)** 0.35 **(d)** 0.45

114. What is the maximum value of $P(B \cap C)$?
(a) 0.1 **(b)** 0.2 **(c)** 0.35 **(d)** 0.45

*Consider the following for the next **two (02)** items that follow:*

An unbiased coin is tossed n times. The probability of getting at least one tail is p and the probability of at least two tails is q and $p - q = \frac{5}{32}$.